ActInf Textbook Group ~ Cohort 3 ~ Meeting 2 (Chapter 1, part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_3 | *Cohort 3 ~ Meeting 2 (Chapter 1, part 1)* | *57:25*

All right, hello everyone. It's February 1st, 2023, and we're in cohort three of the Paradol
textbook group, meeting number two. We're having our first discussion on chapter one. So
would anybody like to just give any opening thoughts or comments,
anything that they expected, preferred, enjoyed, remembered about chapter one? Like what was
something they were expecting and got or didn't, or something that stuck with them
about the experience of chapter one? You can raise your hand or just go for it.
How about, oh, yes, please go for it, someone, or chance favors the prepared mind.
Go for it.
Awesome. Yes, thank you. And I'll copy your comment in, Jonathan.
Yes, rereading, taking another look, and just letting it play out the first read and then
coming back again and again and seeing more each time. And a lot of the formalisms are not in chapter
one. So there is meat slash nutrition elsewhere. But also there's interesting insights within the
chapter, and I'm sure it sparked people to think about different things. So anyone else, feel free
to go for it. Like just any sections of chapter one or any framings that you remembered or you highlighted
or that stuck out to you?
John Luigi? Oh, thank you.
John Luigi? Oh, thank you.
In terms of what motivates or excites people about learning ACT-INF, they situate the question that ACT-INF
seeks to address as how do living organisms persist while engaging in adaptive exchanges with their
environment. Pretty interesting. Is that the question that people thought active inference
was going to seek to address? Do they think ACT-INF may address another question as well?
Okay. As always, just like feel free to raise your hand or write something in the chat, or we'll just
keep exploring along. So I've not come across the neat Scruffy's dichotomy, so that was a nice
perspective. Let's look at that. Does anybody want to
summarize the neat Scruffy's dichotomy? Or give like what did they think neat Scruffy's
means? Or what was it doing at this part of the book?
And I think my take on it, so I'm a physicist originally, and so looking at it from sort of
building from the ground up, you know, creating unified theories of things where everything is
explainable within some framework versus those who try to look at the details and build from the
details up to understand the phenomena. That was kind of my perspective on this.
All right. Ali, and then anyone else?
Yeah, well, actually, I recommend this essay each time because I really like it.
In the book, Andy Clark and his critics, there's a beautiful essay by Karl Pristin, namely Beyond the
Desert Landscape, in which he explains in a very beautiful prose the difference between the low road
and the high road and the high road. And it goes into much further details about the significance of

these two approaches and the pros and cons of which and also the importance of looking at the same phenomena, call it intelligence, sentience, or agency or whatever, from these two perspectives. And what can we glean from looking at it from those two perspectives. So I highly recommend reading this essay, which you can find a PDF for that essay on resources page as well.

Thanks. Thank you.

Thank you.

High road, neat, unified imperatives, scale-free systems. Low road gets scruffy. There's a lot of last miles in the low road because every generative model is different. And they're complementary roads to meet in the middle with active. Kate, or anyone else?

This is still pertaining to the scruffies and needs. I was just, it just occurred to me as I was both reading the introduction and now listening to this, how many of us here would be self-described needs and self-described scruffies. I'm personally definitely a neat. I'm very much gravitating towards an underlying common explanation for everything. So which draws me to the active inference as such. So I'm just wondering in the group of us, where would we fall in that? Just out of curiosity.

Awesome.

Very fun question. Yeah. Does anybody want to dox themselves in terms of being a neat or a scruffy or some secret third thing and just share like why that resonates with them and how they might see learning active bolstering how they are or providing some other support blue and then anyone else?

So definitely a scruffy.

And, you know, like, I mean, I strive towards neat, right? Because I realize that like order is sometimes preferable. I'm drawn to active inference because I really feel like it's a kind of a creative lens with which to view like everything. So it's just another like set of rose-colored glasses that I put on.

And like I also paint and do other things. And so sometimes like I'll be staring at a person like, I wonder how I would represent their nose in charcoal. Like so just it's like one of those things. It's just a pair of glasses that I put on. But I do want to also draw the connection between what we were talking about yesterday on the book stream. So I don't know who's watching that. But we had like, you know, they're talking about leadership styles. And there's like the antagonistic neural network perspective where there are two like opposing, not opposing, maybe unifying neural networks. One is called the task positive network. And the other is the default mode network. With the task positive network being very like oriented at getting things done. It struck me as like a more orderly thing, like maybe in line with like the needs and the direct or default mode network being more like people centered and creativity focused. And like, it's also like the difference between like right and left brain. I mean, we hear all these things all the time, right? There's like the logical versus emotional. And so perhaps like, it's maybe not neat or scruffy, but maybe it's both, right? I don't know.

Thanks, Ali, and then anyone else?

I also think that this distinction between needs and scruffies somehow relates to reductionism versus emergentism, which is pretty hot debate nowadays in the philosophy of science. So, well, of course, we know that science, for the most part, was basically a reductive endeavor and most scientists by large are reductivists. But nowadays, especially from the mid 90s, and the advent of the complexity theories and dynamical systems and so on. Another viewpoint called emergentism is actually emerging, no pun intended. And, but there was always a debate between which one can provide the most insightful discussions in, in each particular situations. But nowadays, some philosophers kind of take the middle ground between those two, and the most important points of view, such as the contextual emergentism, which has been described in this recent book, Emergence in Context. And I think active inference kind of maps, in my view, perfectly to this point of view of contextual emergentism. And it's the best of two worlds. In some sense, it's a purely reductive viewpoint. And in another sense, it's it can be described as an emergentist viewpoint, at least the branch of emergentism called weak emergentism.

So, Maria, I fail to see how one can work with details without some theory behind.

So, Maria, I fail to see how one can work with this. And I'm going to see how one can work with this.

So, I'm going to see how one can work with this. And I'm going to see how one can work with this.

Anyway, I can give a thought. But I totally agree.

Implicitly, always, and explicitly, often. We do need some sort of combination.

And also, it kind of maps just like even a little bit more loosely, like industrial versus farmer's market or boutique.

And I think in the book, they basically say like, right before the needs and scruffies come.

Okay. Here's the two perspectives.

One perspective is that just the incredible diversity of biological systems and neural processes each require a dedicated explanation.

So, we're going to have a journal or a theory or an acronym or an undergraduate major for just serotonergic synapses.

That's one option. Every single phenomena is going to have its own farmer's market, bespoke, scruffy theory.

That would lead to the proliferation of fields with little hope for unification.

The neat perspective is recognizing diverse manifestations,

but asking whether those phenomena of broadly perception, cognition, and action from a lens of persistence and adaptation in life,

whether there might be some coherent explanations that, again, recognize the diversity of manifestations and then help us understand it.

Michael and then Jonathan.

Michael or Jonathan.

I'm sorry.

Michael.

Go ahead, Jonathan. I'll follow.

Just to say that perhaps this sort of dichotomy between a high road and a low road is kind of a matter of perspective

in the sense that active inference itself, you know, it's a description of certain types of systems,
but there are other types of systems out there.
And so, if one wants to look at a higher level view, then in some sense, you know,
active inference and the free energy principle are themselves the low road of some higher level principle,
high level physics principle, whatever it is you want.
That sounds very interesting.
That sounds very interesting.
roads all the way up slash down uh Michael this in this conversation I may be saying that I'm I
thought I was uh neat and maybe I'm a scruffy uh the uh I want to jump back for just a second to
what is the key question instead of um I I was troubled by that particular phrasing and I I was
like what what would I have possibly said and I'll put in the chat but for me the question is how does
life interrogate life um uh it it there's this dance between poetry and plumbing and that um there's a
lot of this conversation that is a plumbing conversation and a part of the way consciousness
uh makes sense and speaks is poetry and um and that that's one of the things that um that is is the
sense here of not not an either or which is a plumbing way of describing something but um uh
opposites and paradoxes uh that are both um simultaneously considered and that um that
uh consciousness supports um and an active inference supports a way of both um problem solving and
contemplating mystery both concurrently and that this is part of the kind of uh there's something
about not just the problem solving but the the the uh the being related in a way that is not um agentic
if you will uh that I think is is missing in in in some of this thank you and that that really ties
in well with what blue brought up like the task positive network is like the getting things done
and the default mode is like the mind wandering and the receptivity and there's just an oscillation
between those two modes in neural systems nathan welcome and yes please anything you want to add
um yeah so I mean you're talking about plumbing and poetry and you just were talking about tp and dm where
we're switching between these two modes of thinking um and before that we're talking about like higher
and lower levels of you know thinking about this whole problem so are we making the claim that like
active inference is like are we making the claim that this is like the appropriate layer of abstraction
to look at this problem because when we usually look at big problems like this we have to pick a layer
of abstraction where we're not losing like a lot of detail and resolution but um by going higher up in
abstraction but obviously we don't want to get bogged down by the complexity of adding more detail and resolution
very interesting question
i guess just one very logistical way to address it is we're reading chapter one that's giving us an overview
of the book then we're going to have like a month to talk about the low road to active inference and the high road
to active inference and that's going to bring us to active inference itself so at least rhetorically
or pedagogically in terms of how to lay out an argument and or lay out an educational

approach it seems like the authors do believe that thinking about neats and scruffies
and the resolution or the the unification of them not through homogenization but by recognizing those two
perspectives that that is an important construct in at the very least understanding and or learning
active and then maybe that will bring us to the question of whether it's also the appropriate level of
abstraction in application ali and then anyone else
uh actually i don't think we can associate a single level of abstraction to active inference framework because
uh as far as i understand it active inference can equally be applied to multiple levels of abstractions and
uh it's somehow uh the nonce it's a scale-free framework or uh in other words where multiple scale framework so
uh even uh if we if we talk about temporal scales or even spatial scales uh we can apply it uh onto various uh
uh various scales and various levels of abstraction so uh yeah it's not uh it's a much more broader framework
uh than we can uh associate with uh some other theories which uh specifically uh are limited to a particular
level of abstraction so um that might we we can see some of the examples for these different levels of
abstractions especially in the second part of the book awesome thanks so um let us turn to some of the
questions that people prepared so does anyone see a question that they wrote that they would like to ask
and we can explore or we'll go with the ones that people have upvoted
but if anybody has like one that they wrote or just want to highlight let's go to it
and also it's awesome like i saw a lot of people adding comments on the questions um
all right let's just go to the most upvoted one
bringing the question down into here so it's just a little more visible
okay on page six it refers to active inference as a normative framework there's some evaluative
standard against which behavior can be scored which is free energy specifically the minimization of
variational version thereof so just to plant the seed but we're gonna come back to it later the two
kinds of free energy that we're going to be talking a lot about are variational and expected but we're
going to come back to that for now let's just think about free energy as a unified imperative for
perception cognition and action variational is in the present and expected has to do with futures
then it refers to planning to achieve distal goals in humans does this definition of normative mean
that active is agnostic about precisely what humans goal that might be one human may wish to
save the world another may wish to rule the world both are essentially minimizing free energy
i might be mixing up a few concepts here but the gist is that active models don't predict a socially
normative course of action would that be correct to say all right what would anybody like to add on
normativity of active what does it mean to say that active is a normative framework does that refer to
social norms or what kind of normativity is being described
ali go for it
i guess at the most fundamental level uh the normativity admits uh the agent uh to be persist through time
uh so uh one meaning of normativity can be uh seen from the most fundamental requirement of a system or
quote-unquote thing uh to actually exist or persist through time but of course we can apply this concept
onto some uh higher levels of uh conceptions such as social normativity and so on but

yeah again it applies to multiple uh levels of abstraction thank you jonathan

yeah that that sort of requirement for wanting to persist through time feels like an evolutionary imperative and there are other imperatives into in terms of our sort of homeostatic set points and comfort and and hunger and things like that which then lead to um the thing which has been evolutionarily um uh sort of chosen through time so it feels like the two kind of have a um there's a balance between the two in in terms of hierarchies of scales of time i guess

thank you thank you scott for bringing that quote

from a zero

active, and linear regression. Those are like the features that differentiate active and linear regression. So we might say that a linear regression is normatively using this kind of imperative function, like to fit the least squares or the L2 norm. And it's just saying this framework is going to use this imperative to draw a straight line through points.

And kind of by analogy, ACDIMPH is going to use an imperative. Turns out it's a free energy functional that we're going to go into in the coming chapters. It uses that unified imperative not to draw a straight line through points like a linear regression does, but to describe Bayes' optimal perception, cognition, and action.

So it's doing something similar. Instead of having like one imperative for perception, and then a different imperative for action, it's a unified model of perception, cognition, and action that we can use this one imperative, the free energy of a generative model, and use that to guide the fine tuning or the learning of that generative model.

To return to the social question as we move forward, no, it is not like an ought. It's not the is ought. This is not an ethical framework. Just because there's a normative way to fit a linear regression doesn't mean that it tells you what any given culture is going to be happy or upset or benefit from or anything like that. There's probably a lot, there's other ways to say it, but just the short answer is this is correct. ACDIMPH is not in the kernel of ACDIMPH is not describing social normativity in the sense of like peer pressure or like what people will look at you weirdly if you do.

Okay.

Next question.

Living organisms can only maintain their bodily integrity by exerting adaptive control over the action perception loop, which they frame that adaptive control as the question that ACDIMPH seeks to address.

Do generative models represent and thereby predict bodily integrity?

Or do these models predict the causes of their sensory inputs?

And via minimizing the prediction error, they achieve maintaining their bodily integrity

without explicitly predicting bodily integrity.

How to best talk about what a generative model is predicting?

Great questions. What does anyone want to add?

Great question.

So

Well, I think one short answer, and then Ali, is that generative models fit whatever parameters they have.

So you could imagine making a generative model where there's a hidden state that is bodily integrity, and then there's observations that help reduce uncertainty about bodily integrity.

So in that case, we'd say yes, the generative model represents bodily integrity as a hidden state, and therefore tries to do inference using that.

Or do these models predict the causes of their sensory input, and via minimizing prediction error, they achieve maintaining their bodily integrity?

That's actually like another way to say something similar, which is that the hidden states are the latent causes of sensory observations.

And so by minimizing the prediction error with respect to observations, it is possible to achieve maintenance or homeostasis of those latent states, even though they are explicitly predicted or represented, but they're not observed.

So

Ali, anyone else?

Well, actually, in active inference, all the agents, or more technically, all the particular systems, and all the particles, sorry, do, are kind of doing the, this kind of inference making tasks, in order to sell the data by

self-evidence and actually this task of self-evidencing is the fundamental

task that an agent needs to undertake in order to keep its homeostasis into action or

doing that kind of homeostasis through allostasis because both of those terms are obviously related

one of which needs more active and more and more active mediation to undertake the homeostasis

task but in any case the concept of self-evidencing exactly refers to this kind of integrity maintaining

a task that any self-reference any inference making agent needs to undertake to persist through time

so i think this is the fundamental requirement of any agents modeled in active inference framework awesome thank you and once we start into the subsequent chapters and we look at how preference

is used in active inference we'll have some awesome discussions like do we expect homeostatic

temperatures or do we prefer them it's like on one hand i prefer it's what i like on the other hand

it's what i expect because i wouldn't be surprised to find myself there so how does that play out um ever

yes yes did you hear me okay okay so it could could an answer also depend on which level of the hierarchy

you are talking um so at the lowest of hierarchy there's sensory uh stimuli that are being predicted

and the more you move up the more abstract models or predictions are taking place so higher up the hierarchy

i could have predictions about bodily integrity um or predictions about my continuing existence

uh etc and then lower down the hierarchy you could have some physiological variable that's predicted like

nociception in the case of pain and then you can have all all

awesome okay was that is that all yes okay yes totally agree um often people talk about like nested models

models or hierarchical models or hierarchical models multi-scale models and that's one of the strengths of

taking of at least bringing some neatness into play is like by using a level of neatness that lets us

generalize across different cognitive phenomena like what we might just conversationally call low level or high level

it allows those models to be composed in an integrated way so in this textbook there's not going to be an emphasis

on nested modeling it's going to focus on like the kernel of active inference which is like the single agent

at the single level but then there's all of these elaborations which are active areas of research and development

um outside of the textbook yes thank you jana

could you clarify low and high level

yeah i think they uh if i think about the cortical hierarchy there's i think six layers

uh and and if i read the literature correctly they talk about uh different kind of predictions depending on

which kind of cortical layer you are talking about

and and the active inference models are built in a way that conf that also fits

um the anatomy of the brain so they talk about computational anatomy fitting the active inference models

thank you jonathan and then gian luigi

um yeah i i think that that's right i think another perspective is is simply that low level is for

instance the the pure sensory inputs themselves um we may be making predictions about you know a site

that um something that will come to our eyes higher level might be some um sort of more abstract idea

of of the thing that it is outside in the world that is is giving us the sensory input and so there's

a sort of hierarchy of um time scales and physical scales um that we might take in moment to moment and

build up a bigger and bigger picture awesome so just to restate that and then gian luigi the low level

this is just one spatial metaphor is that the pure sensory inputs themselves like the low level is where

the rubber hits the road and then the higher levels are abstract or synthesized representations

about what are giving rise to those sensory data and that is also called a cause in the statistical sense

HIM HIM

like those cortical layers or the the lower levels are the most most um low level sensory uh inputs

so so so to say and the higher levels are the more abstract ones but i do believe that they also

uh that the distinction might also not be that strict

all right and then ever and there's a few more points to make

yeah and this question uh about what is predicted also

yeah um i think it also has to be uh seen from this uh instrumentalist and realist discussion

uh is it is it the modeler that uh is building an active influence model and tries to infer what

the organism at whatever level is predicting uh and and that's a

what i wonder is um you you also read in the literature a lot of times that they talk

with uh the terms as if it looks as if the organism is minimizing this or that or predicting this or that

so this as if language um also makes me wonder how how do we know what an organism is predicting

awesome this instrumentalist and realist and map territory we're going to return to a lot

um on i want to bring a few more points on the nested models and then we'll return to the questions since

um like nested modeling it's kind of like the active kernels like one lego brick

with like a bottom and a top part and then like it is very amenable and as soon as we have the legos

we want to stack and put them lateral and build them but the multi-level lego structure is predicated on what that one is

at least understanding what it is so in live stream 28 on um sandved smith's paper this is just one example

of a nested model with three layers where the quote low level is sensory and then there's a level of

attention and meta awareness so there's not just one way to do nested models there's many nested models

that one can make but it's an example of how something that is like raw and exteroceptive

can be seen as a low level model nested within

other models that are quote higher but again that's just like a that's just like a

cultural visual graphical layout there's no reason why it couldn't be from left to right or

top to bottom or inside to out so it's like it's about what it is not just the spatial layout but that's what it is

on the six levels of the brain so here's from um live stream 43 which was with maria

um yes there are six anatomical layers in parts of the mammal brain six layers that are defined like

histologically like the tissues look different but it's really important to remember that is not the same

thing as it's not six nested legos and we're going to get to this more when we get to chapter five which

is all about um the neuroscience but here are those six layers so this is not six full nested models
like here's three full nested models that are truly nested whereas the six histological layers
actually are part of an integrated inference scheme it's not six nested models
but a lot of times people think oh multi-level models hierarchical bayesian modeling
hierarchical predictive coding architectures six layers in the brain six nested models that's not true
but that's a very common um framing
so what is the generative model predicting whatever it is set up to predict these are maps not territories
okay i think okay great there was some some awesome discourse here so can someone explain to me
why the inferential problem is intractable page eight it has to do with the marginal likelihood p of y in the
bayes the theorem so does anyone here who is in this discussion like want to summarize it or what their
understanding of what is the tractability issue and how are we going to approach that um yeah so my the
the sort of example i gave there was that if you see a flash of light you want to know what's the probability
that it is a particular thing given that you have you've you've had some observation of the world um
but in order to do that you need to sum up um in this case all of the different things that could
have given rise to that particular flash of lights and that in itself is in is intractable so there are
various terms within the um within the bayes equation which in order to do it properly in a complex world
would mean summing over an infinite number of things that was my my thought there great great
so there's a lot of ways to to talk about this but you pointed to that that several pieces are required
to go from a stimuli to inference about a causal state of the world like whether it was truly a tiger
giving rise to the stimuli or whether it's something different you'd want to know a few things you'd
want to know the probability of given that it's a tiger what is the probability it would look like
that flash you saw but then also you want to know how likely is that flash from other possible hidden
causes and so that is like the probability the flash given that it's a tiger or given
it's a chipmunk given it's all these other things and that can be an open-ended sum
so thanks for that great answer it is in essence an infinite sum of possible things
and you'd have to know the probability to be able to say what you should really think

will be able to say that with a HIM

a sentence, the marginalization operation might require analytically intractable integrals.
And I don't understand what that means.
Is it different than the answer you gave before?

It's very related.
So in a discrete setting, we are talking about sums.
Like sums, you know, 1, 2, 3, 4, 5.
Sum over 1, 2, 3, 4, 5.
In a continuous setting, we are talking about integrals.
So if something is being summed over a discrete range, that is analogous to it having an integral over a continuous range.
And so they're both beset with the same challenges of large and interacting spaces.
And then just one kind of note, because on a Bayesian and marginalization, this is like actually the margins of a piece of paper.
That's what it's referring to when there's kind of like actuarial tables back when statistics was done on spreadsheets.
So it's like if you have states of the world and then observations in a grid.
If each observation mapped onto one hidden state of the world, then you would just have like only on the diagonal would you have any numbers.
We're going to come back to these topics.
But then you can imagine that you might be interested in marginalizing a given row, like summing across that row.
So saying, okay, how likely is this across all columns?
And that's called marginalization.
So why is it intractable?
Because it might just be requiring computational resources that are unrealistic or implausible.
I think there was actually the whole reason for a variational free energy quantity.
This marginal likelihood thing.
Is that correct?
Yes, it is.
Free energy is going to help bound our surprise.
Let's come right to it.
So this one kind of goes in multiple different directions.
The central imperative is that organisms maintain their existence.
High road.
Avoid surprising states.
How can one explain different self-harms or perceived self-harms?
This is kind of like the social norms question in that we're not always going to get a neat answer from a human scale complex nested system where the generative model hasn't been stated.
If that makes sense.
So like given a generative model, one could describe how driving off a cliff is the generative model's most likely course of action.

Like if the generative model is I always drive straight, I'm likely to find myself driving straight.
That generative model may drive off a cliff.
But to kind of not mention the generative model and just say, well, how is it possible that surprise is bounded by driving off cliffs is not a full, it's not fully making contact with what the surprise is because surprise is with respect to a generative model.
So whether it's the extremely complex cases of humans or whether we're talking about something different, then surprise and the bounding of surprise through free energy is with respect to a generative model.
Everything is always in respect to generative model.
There's no surprise minimization, free energy minimization outside of a generative model.
So surprise minimization, free energy minimization, those are qualities of a generative model.
Just like the least squares regression, the sum of squares is with respect to a linear regression model.
Surprise bounding free energy is with respect to an active inference generative model.
And once the generative model is specified, then one may find different actions as avoiding surprising states, including addiction and different states that people have studied that we're going to come to, but we need to fill out what generative model we're talking about before applying the imperative.
Jonathan and then Everett.
Yeah, within that generative model, there are also the different desired states that the organism might want to be in.
And those desired states, they're not normative in the sort of sociological sense.
We may have different desired states.
And so for some people that may be very different from what another person wants and may lead to one of these outcomes.
Thank you.
And Everett.
Yes, this example with the suicide is also related to the high roads where the imperative is to maintain our existence, minimizing surprise necessary for survival.
So if I commit suicide, then I would violate that core imperative.
But we do know that people commit suicide.
So how can we explain that?
So this framework.
And it also has to do with if I predict bodily integrity.
But for example, I am someone who experiences chronic pain, because that's the best explanation for my prediction error that I have will be an update of the model that says, okay, now I'm not predicting any bodily integrity anymore, but threats to my bodily integrity.

And would that also then lead to eventually committing suicide or just some thoughts?

But I can't make sense of this.

Thanks.

Okay.

Just in our final minutes to touch on one or two more, all these discussions, people can continue to engage and develop, add resources, ask more questions.

Like it doesn't stop here.

This is like the tip of the iceberg of us wanting to develop the state on these questions, because there are things that we're all asking.

There's things that other people are going to be curious about.

And we can have great answers and complex discourse and multiple perspectives here.

We're just going to do one more, and then we'll end two minutes before the hour.

And again, these are also things we're going to return to in future weeks.

This is just our first week on chapter one.

Next week, we're going to do another chapter one session, and we'll come through more questions, the second half of the questions on chapter one, and anything else that people add and upvote.

So just in the last three minutes, can someone explain why exploration and exploitation are automatically balanced through policy selection in active inference?

Page 10.

What does it mean to say both exploration and exploitation are balanced?

Jana, thank you.

I will save this chat.

Thank you.

Thank you.
In my understanding, it would be that you only explore as much as necessary to exploit.
So exploration is an effort somehow, and you want to minimize this to get the maximum of exploitation.
That's my understanding approximately.
Thanks.
One very hands-on way to explore this question is in Model Stream 7.2 from just a few days ago.
In the video description of Model Stream 7.2, there is a notebook that can be run in your browser without any more tweaking.

And it is going to have an agent that has the choice to do a few different things.
They can play either of two different slot machines, one of which is better than the other, or they can get a hint.
And so to have an imbalanced strategy would be like to kind of zone into the first slot machine you go to and stay there.
To have an overly imbalanced strategy towards exploration, that might look like just getting the hint every time or just kind of moving around really rapidly in the space, but not ever like latching on to ones that have high utility.
And then a balanced strategy, which there's not just one single answer to this, but a balanced strategy in a given situation is an action policy selection.
It's a strategy or tactics that navigate or manage the tensions of the requirements between exploration and exploitation.
And sometimes that's modeled with what's called a multi-armed bandit, where there's like a bunch of slot machines, you want to win the most money, but you don't know how good each of the slot machines are.
So you got to explore a little bit to find out like how good they are, but then you want to spend most of your time on the best ones.
You want to spend time proportional to how good they are.
But of course, if you only spend time on what you perceive as the best, things might be changing.
And so you might end up not being on the best machine soon.
So thank you everybody for joining.
We'll come back next week for the second discussion on chapter one.
And please add comments and questions and discourse in the time till then.
But thanks everybody for the awesome discussion.
Farewell.
Thanks everyone.